

Ivan Ilin

Machine Learning Researcher | PhD Candidate in Computer Science at KAUST

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Research Profile

Machine learning researcher developing efficient optimization and compression methods for large language models. My work spans pruning, quantization, sparse fine-tuning, pipeline parallelism, and distributed optimization, with an emphasis on methods that connect rigorous analysis to practical open-source systems.

Education

King Abdullah University of Science and Technology (KAUST) Thuwal, Saudi Arabia
Ph.D. in Computer Science Jan 2025–Present

- Advisor: Peter Richtárik. Advanced to PhD candidacy after successfully defending the proposal *Efficient Optimization and Compression Methods for Large Language Models*, July 2026.

King Abdullah University of Science and Technology (KAUST) Thuwal, Saudi Arabia
M.S. in Computer Science 2023–2024

- Thesis: *Efficient and Fast Pruning of Large Language Models*. Advisor: Peter Richtárik. DOI: 10.25781/KAUST-44481.

Novosibirsk State University Novosibirsk, Russia
M.Sc. in Automation of Physical and Technical Research 2021–2023

GPA: 5.0/5.0.

Novosibirsk State University Novosibirsk, Russia
B.S. in Automation of Physical and Technical Research 2017–2021

GPA: 4.67/5.0.

Research and Professional Experience

Optimization and Machine Learning Laboratory, KAUST Thuwal, Saudi Arabia
PhD Researcher Jan 2025–Present

- Develop optimization and compression methods for large language models, including block-wise pruning, sparse fine-tuning, and pipeline-parallel training.
- Release reproducible implementations and collaborate on research in LLM quantization and asynchronous distributed optimization.

ExpaSoft Novosibirsk, Russia
Deep Learning Junior Researcher 2020–2023

- Accelerated a Tacotron 2/WaveGlow speech-generation pipeline with structured pruning and distillation; achieved more than 1.5x speedup with improved quality, or 2x speedup with limited quality loss, on an NVIDIA GTX 1050 Ti.

Budker Institute of Nuclear Physics Novosibirsk, Russia
Undergraduate Research Assistant Jan–Aug 2020

- Built automated tests for Belle II electromagnetic-calorimeter electronics and measured 63% ECL library coverage with gcov.

Grants and Distinctions

KAUST Small Translational Research Grant (sTRG) \$100,000
TRACE Project 2026–2027

Transforming Textbooks into Verified, Interactive Courses using Retrieval-Augmented Generation. Primary author of the successful proposal and project team member; lead PI: Peter Richtárik. Award period: July 2026–June 2027.

NeurIPS 2024 Oral: *PV-Tuning: Beyond Straight-Through Estimation for Extreme LLM Compression*.

Teaching Experience

King Abdullah University of Science and Technology Thuwal, Saudi Arabia
Teaching Assistant, CS 260: Design and Analysis of Algorithms Fall 2025

Novosibirsk State University Novosibirsk, Russia
Programming Teacher (Python) 2022–2023

Peer-Reviewed Publications

- [1] Vladimir Malinovskii, Andrei Panferov, **Ivan Ilin**, Han Guo, Peter Richtárik, and Dan Alistarh. *HIGGS: Pushing the Limits of Large Language Model Quantization via the Linearity Theorem*. NAACL 2025 (Long Papers), pp. 10857–10886. [paper] [doi]
- [2] Vladimir Malinovskii, Denis Mazur, **Ivan Ilin**, Denis Kuznedelev, Konstantin Burlachenko, Kai Yi, Dan Alistarh, and Peter Richtárik. *PV-Tuning: Beyond Straight-Through Estimation for Extreme LLM Compression*. NeurIPS 2024 (Oral). [paper] [oral session]
- [3] Alexander Tyurin, Marta Pozzi, **Ivan Ilin**, and Peter Richtárik. *Shadowheart SGD: Distributed Asynchronous SGD with Optimal Time Complexity Under Arbitrary Computation and Communication Heterogeneity*. NeurIPS 2024. [paper]
- [4] Jihao Xin, **Ivan Ilin**, Shunkang Zhang, Marco Canini, and Peter Richtárik. *Kimad: Adaptive Gradient Compression with Bandwidth Awareness*. 4th Workshop on Distributed Machine Learning (DistributedML 2023), pp. 35–48. [paper]

Preprints and Technical Reports

- [1] **Ivan Ilin**, Philip Zmushko, and Peter Richtárik. *Super-Tuning: From Activation-Aware Pruning to Sparse Fine-Tuning*. arXiv:2607.09287, 2026. [paper] [code]
- [2] **Ivan Ilin** and Peter Richtárik. *Demystifying Pipeline Parallelism: First Theory for PipeDream*. arXiv:2606.03498, 2026. [paper] [code]
- [3] Abdurakhmon Sadiev, Artavazd Maranjyan, **Ivan Ilin**, and Peter Richtárik. *Ringmaster LMO: Asynchronous Linear Minimization Oracle Momentum Method*. arXiv:2605.18174, 2026. [paper] [code]
- [4] **Ivan Ilin** and Peter Richtárik. *Thanos: A Block-wise Pruning Algorithm for Efficient Large Language Model Compression*. arXiv:2504.05346, 2025. [paper] [code]
- [5] **Ivan Ilin**. *Hessian of Perplexity for Large Language Models by PyTorch autograd (Open Source)*. arXiv:2504.04520, 2025. [paper]

Selected Research Presentations

PhD Proposal Defense, KAUST <i>Research proposal defense</i> <i>Efficient Optimization and Compression Methods for Large Language Models.</i>	Thuwal, Saudi Arabia July 2026
KAUST Rising Stars in AI Symposium 2025 <i>Poster presentation</i> <i>Thanos: A Block-wise Pruning Algorithm for Efficient Large Language Model Compression.</i>	Thuwal, Saudi Arabia April 2025
SANDS Seminar Series, KAUST <i>Research seminar</i> <i>Thanos: A Block-wise Pruning Algorithm for Efficient Large Language Model Compression.</i>	Thuwal, Saudi Arabia Nov 2024

Selected Open-Source Software and Outreach

VoiceCut <i>Creator and Maintainer</i> Local-first audio and video editor that uses machine learning to identify and remove failed takes, repetitions, false starts, and breaths.	Open source 2025–Present
Thanos, SuperTuning, and Randomized PipeDream <i>Research Software</i> Reproducible implementations for LLM pruning, sparse fine-tuning, and pipeline-parallel optimization.	Open source 2024–Present

Technical Skills

Languages and frameworks: Python, C, C++, PyTorch. **Tools:** Git, Linux/Unix, LaTeX, OpenCV, OpenGL.
Methods: Large language model compression and evaluation; distributed optimization and training.